

CURRICULUM VITAE OF JOÃO MIGUEL MACHADO

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RESEARCH INTERESTS

I am interested in the Calculus of Variations, with an emphasis on variational problems at the interface between optimal transport and geometric measure theory, as well as their applications to measure quantization and the quantitative stability of variational problems and functional inequalities.

PROFESSIONAL EXPERIENCE & EDUCATION

Since October 2024: *Postdoctoral Researcher in the Optimal Transport group supervised by Guillaume Carlier, Alessio Figalli, Quentin Mérigot, and Filippo Santambrogio*
Institut Louis Bachelier, LMCRC, Paris.

September 2021 to October 2024: *PhD under the supervision of Antonin Chambolle and Vincent Duval. Thesis entitled “Optimization in spaces of measures: Optimal transport, geometric structures and game theory”. [Link](#)*
Université Paris-Dauphine and Inria Mokaplan team, Paris
Funded by the SDOSE doctoral school fellowship.

September 2020 to September 2021: *Master’s degree (M2) in Optimization*
Université Paris-Saclay and École Polytechnique
Funded by the Sophie Germain program of FMJH and internship at INRIA Paris.

2016 to September 2020: *I worked in Brazil under the supervision of Professor Maria Soledad Aronna during my undergraduate studies in electrical engineering. This led to an internship at Inria Saclay, in the former COMANDS team, under the supervision of Frédéric Bonnans, and subsequently to a master’s degree awarded by FGV.*
Fundação Getulio Vargas, Rio de Janeiro.

2014 to September 2018: *Engineering School*
IME (Military Institute of Engineering), Rio de Janeiro, Brazil.

PUBLICATIONS AND PREPRINTS

9. **Quantitative Stability for Minkowski’s problem** (joint work with Károly Böröczky and João P. G. Ramos). arXiv:2603.17726 ([Link](#)).
8. **On the Wasserstein gradient flow of semi-discrete energies: evolution of urban areas and uniform quantization.** hal-05535338 ([Link](#)).
7. **Quantitative stability for the Brascamp-Lieb inequality and moment measures** (joint work with João P.G. Ramos). 2025. arXiv:2511.22636 ([Link](#))

¹Updated March 19, 2026

6. **Absence of loops for the Wasserstein- $\mathcal{H}^1$ problem: the concentration/blow-up argument.** 2025. ([Link](#))
5. **From Nash to Cournot-Nash via Γ -convergence** (joint work with Guilherme Mazanti and Laurent Pfeiffer). 2025. ([Link](#))
4. **Phase-field approximation for 1-dimensional shape optimization problems.** In *SIAM Journal of Mathematical Analysis*, 2025. ([Link](#))
3. **One dimensional approximation of measures in Wasserstein distance** (joint work with Antonin Chambolle and Vincent Duval). In *Journal de l'École Polytechnique-Mathématiques*, 2025. ([Link](#))
2. **The Total Variation-Wasserstein problem: A new derivation of the Euler-Lagrange equations** (joint work with Antonin Chambolle and Vincent Duval). In *International Conference on Geometric Science of Information*, 2023. ([Link](#))
1. **The shooting algorithm for partially control-affine problems with application to an SIRS epidemiological model** (joint work with Maria Soledad Aronna). . ([Link](#))

AWARDS AND DISTINCTIONS

- IMPA 2026 summer postdoctoral fellowship: *Fellowship for a postdoctoral research stay at IMPA*
- EMap (FGV-Rio) Summer school lecturer: *Invitation to give a 30-hour summer course on optimal transport at EMap (FGV)*, [Link](#)
- Dodu Prize – SMAI MODE Days 2024
- Best Poster Award – LACIAM 2023 (Latin American Congress on Applied and Industrial Mathematics)
- Sophie Germain Fellowship 2020 – FMJH (Fondation des Mathématiques Jacques Hadamard)

TEACHING EXPERIENCE

- February–May 2026: Teaching assistant at NYU Paris, course “Linear and Nonlinear Optimization” taught by Alfred Galichon.
- January–February 2026: Instructor for a 30-hour summer course on optimal transport theory and applications, FGV-EMap, Rio de Janeiro, Brazil. <https://emap.fgv.br/programa-de-verao#cursos-e-ementas>
- During my PhD and studies in Brazil, I participated in several teaching activities. A summary of my teaching experience follows:
 - Teaching assistant (L2–L3), Université Paris-Dauphine:
 1. 2024: Teaching assistant and lab instructor for “Numerical Methods: Matrix Algebra” and “Functional Analysis”
 2. 2023: Teaching assistant and lab instructor for “Numerical Methods: Matrix Algebra” and “Numerical Methods: Optimization”
 3. 2022: Teaching assistant and lab instructor for “Numerical Methods: Optimization”
 - 2017–2020: Teaching assistant in several courses at EMap, School of Applied Mathematics, FGV-Rio. Courses included:
 - * Functional Analysis (doctoral program, 2020), Prof. Maria Soledad Aronna
 - * Measure Theory (doctoral program, 2020), Prof. Vincent Guigues
 - * Stochastic Processes (undergraduate, 2019), Prof. Yuri Saporito
 - * Real Analysis (master’s level, 2019, 2017), Prof. Maria Soledad Aronna
 - * Real Analysis (undergraduate, 2019), Prof. Maria Soledad Aronna

SERVICE AND SCIENTIFIC EVENT ORGANIZATION

1. Referee activity for the journal “Advances in Mathematics”
2. Co-organizer of “GT CalVa”, the monthly Calculus of Variations seminar of the Paris region, since September 2025, <https://indico.math.cnrs.fr/category/424>
3. Co-organizer of the “Mokameeting”, the monthly seminar of the MOKAPLAN team, September 2023–September 2024
4. Member of the local organizing committee of the “Workshop in Optimal Control and Mean Field Games”, Rio de Janeiro, October 2019, <https://sites.google.com/view/control-mfg-rio2019/home>

RESEARCH TALKS AND PRESENTATIONS

1. Analysis Seminar, IMPA, Rio de Janeiro, January 2026
2. Talk at GT CalVa (Paris Calculus of Variations seminar), Paris, 2025
3. Talk at the minisymposium “Calculus of Variations and Geometry II”, 12th SMAI Biennial Conference, 2025
4. Talk at the minisymposium “PDEs and Applications in Biology and Population Dynamics”, SMAI 2025
5. Contributed talk, SMAI-MODE Days, March 2024 (**Dodu Prize**)
6. CEREMADE PhD seminar, March 2024
7. OT-PDE-ML Working Group Seminar, LMO, December 2023
8. Seminar, University of Freiburg, Germany, November 2023
9. Talk and proceedings paper at GSI’23 (Geometric Science of Information), August 2023, Saint-Malo
10. Contributed talk, “Calculus of Variations in Paris” conference, June 2023
11. Poster presentation, LACIAM 2023 (**Best Poster Award**)
12. CEREMADE PhD seminar, December 2021

LANGUAGES

- English: fluent
- French: fluent
- Portuguese: native language